

Model Development Document

Discrete-Time Logistic Hazard Model for
IFRS 9 Mortgage Probability of Default

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This document describes the development of a PD model for retail mortgage portfolios under the IFRS 9 expected credit loss framework.

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1. Introduction

Under IFRS 9, financial institutions must hold loan loss provisions based on forward-looking estimates of expected credit loss. The central input to this calculation is the probability of default, the likelihood that a borrower fails to meet contractual payment obligations within a specified horizon. IFRS 9 requires PD estimates to be conditioned on macroeconomic forecasts and updated as conditions evolve, which makes the modelling problem fundamentally dynamic rather than through the cycle.

This document describes an end to end credit risk framework for a retail mortgage portfolio, estimated on loan level performance data from the Fannie Mae Single Family dataset. The framework produces 12 month conditional PD, lifetime PD, and portfolio level ECL under three macroeconomic scenarios. It has three components: a discrete time logistic hazard model for loan level PD estimation, a Bayesian VAR for generating forward looking macro scenario paths, and a survival weighted projection engine that combines the two to produce lifetime ECL.

Two features of the data and the problem required explicit methodological choices. First, the raw Fannie Mae archive exceeds 57 GB. Processing it in full is tractable but naive estimation on the resulting panel produces biased results when defaults are rare. Monthly performance observations are aggregated to quarterly frequency, which both reduces computational burden and raises the per row default rate by consolidating sparse monthly default flags into quarterly events. Even after aggregation, quarterly default rates outside stress periods run well below 0.5%, which flattens the likelihood surface and inflates coefficient variance under standard MLE. We address this through a case control sampling design with a weighted exogenous sampling maximum likelihood correction applied to the model intercept. Second, IFRS 9 lifetime ECL requires forward looking macro paths rather than point estimates. The framework generates these through a BVAR with structural long run anchors calibrated to NAIRU, potential growth, and the neutral rate.

The estimation sample covers origination cohorts from 2000Q1 through 2011Q4, with performance observed through 2025. The model is trained on observations through 2019Q4 and tested on the remaining out of time window, which includes the 2020 to 2021 COVID forbearance period. Forbearance provisions allowed borrowers to pause payments without triggering delinquency flags. What this means for the test sample is that defaults which would ordinarily have been observed during this window were suppressed, compressing measured discrimination. The train test Gini gap should be read in that context. The final hazard panel contains 54 million loan quarter observations across 2.7 million loans.

2. Data

2.1 Source and Portfolio Scope

The analysis uses the Fannie Mae Single Family Loan Performance dataset, which provides monthly observations for residential mortgages originated since 2000. Each observation records the loan’s delinquency status, unpaid principal balance, and a set of borrower and loan characteristics. The dataset covers prime conforming mortgages only: full documentation, conventional financing, fixed rate, single family, fully amortizing. Adjustable rate mortgages, interest only loans, balloon mortgages, government insured loans, and loans with prepayment penalties are excluded at source, as are Alt-A loans, reduced documentation originations, and loans with LTV above 97%. What this means in practice is that the estimation sample reflects a relatively homogeneous credit population operating under standard underwriting guidelines, which bounds the generalisability of the model to portfolios with similar risk profiles.

2.2 Cohort Selection and Observation Window

The estimation sample draws on origination cohorts from 2000Q1 through 2011Q4. This window spans a full credit cycle: the pre-crisis expansion, the Global Financial Crisis and its immediate aftermath, and the early recovery through 2011. Loans originated across this range lived through a wide variety of macroeconomic environments, which matters for identifying and validating the macro covariates that enter the PD model. Performance data runs through 2025. The raw dataset contains 535 million loan month observations across 57 GB of source files.

2.3 Aggregation to Quarterly Frequency

Monthly observations are aggregated to quarterly frequency before estimation. The raw monthly delinquency flags are noisy: a loan that misses one payment and cures the following month is technically delinquent for 30 days but is not heading toward default. At monthly frequency these transient events are indistinguishable from the early stages of a genuine default trajectory. Moving to quarterly frequency filters most of this out, because a loan that cures within the quarter never reaches the 90 day threshold that defines default. It also raises the per row default rate since three monthly observations collapse into one row, which reduces the class imbalance before sampling is applied. And because the macroeconomic covariates are reported quarterly, aggregation means the outcome variable and the regressors sit on the same time grid without any interpolation.

Default is defined as the loan reaching 90 or more days past due for the first time. For each loan quarter, delinquency status is set to the maximum across the three constituent

months. A loan quarter is flagged as a default event if that maximum reaches 90 days past due for the first time. Covariate values are taken from the end of quarter observation.

2.4 Hazard Panel Construction

The estimation sample is organised as a discrete time hazard panel. Each row represents a loan quarter observation (i, t) , and the dependent variable is an indicator equal to one if loan i defaults in quarter $t + 1$, conditional on survival through quarter t . A loan exits the risk set at the quarter of first default, so all observations from the default quarter onward are dropped. The row immediately before the default quarter carries a target of one; all earlier rows carry a target of zero.

Each row therefore answers the same conditional question: given that the loan is still performing at the end of quarter t , what is the probability it defaults next quarter? That framing is what makes the discrete time hazard model the right framework here, and it is what allows the one quarter ahead conditional PD estimates to be chained into a lifetime projection in Section 7.

The final panel contains 54,199,536 loan quarter observations across 2,693,621 loans. The train cutoff is 2019Q4. The out of time test window opens at 2020Q1 and runs through the end of the observation period, COVID forbearance window included. The effective non-event sampling fraction τ_{eff} is computed on the full sample prior to this split and carried forward for the intercept correction described in Section 4.

2.5 Variable Construction

Borrower and Loan Characteristics

Four borrower and loan characteristics enter the model and are kept fixed at origination. The rationale for using origination values rather than time varying updates is that lifetime PD projection requires every model input to be either mechanically advanceable or externally forecastable. Macro conditions are handled by the BVAR and loan age advances one quarter at a time, but projecting LTV or DTI over a 30 year mortgage life would require forecasting property values and borrower income paths at the individual loan level, which is not feasible. Origination values then become a natural solution: they are observable, stable over time, and capture the credit quality of the loan as it was underwritten. The variables themselves are chosen because they cover the three dimensions that the mortgage default literature consistently identifies as predictive: equity position, repayment capacity, and borrower credit quality. The consequences of holding LTV static rather than updating it with the HPI scenario paths are discussed in Section 8.

Loan to value ratio (LTV). LTV at origination measures the borrower’s initial equity posi-

tion and captures the proximity to negative equity under a house price decline. Borrowers who originate close to full value have little buffer before the loan becomes underwater, which both weakens their ability to refinance out of distress and reduces the financial incentive to keep servicing.

Debt to income ratio (DTI). DTI at origination measures how much of the borrower’s income is already committed to debt service at the time of taking on the mortgage. It determines how much cash flow remains to absorb an income shock before the mortgage payment is missed.

Credit score (FICO). FICO at origination captures the borrower’s payment history, credit utilisation, and prior delinquency, all of which reflect behaviours that predict future default. It is fixed at origination rather than updated quarterly because projecting future credit scores over a loan’s lifetime is not feasible, and origination FICO is stable enough to remain informative at long horizons.

Loan age. Default rates on mortgages are not constant over the loan life. They tend to rise through the first few years as higher risk borrowers encounter payment difficulties, peak somewhere in the two to five year window, then decline as the surviving pool becomes progressively cleaner. Loan age is included to control for this pattern and prevent the model from conflating duration effects with vintage quality. It enters as categorical buckets: 0 to 6 quarters, 6 to 12, 12 to 24, 24 to 60, 60 to 120, and 120 or more.

Macroeconomic Covariates

Three macroeconomic variables enter the model, selected through the systematic procedure described in Section 3. Their construction and economic rationale are covered there.

3. Macro Variable Selection

Macroeconomic conditions affect mortgage default through several distinct channels: labour market deterioration reduces borrowers’ ability to make payments, falling house prices erode equity and create conditions for strategic default, and broad economic weakness amplifies both. The PD model needs macro variables that capture these channels, are stationary, carry the correct economic sign, and add discriminatory power beyond what loan characteristics alone provide. We run a systematic combinatorial search over a curated universe of candidate series and their transforms, where economic theory determines which series are in the candidate set and what sign each must carry, and the data picks which specific transform and lag performs best.

3.1 Candidate Universe

Five canonical series cover the primary channels through which aggregate conditions drive mortgage default.

Unemployment rate. Rising unemployment increases the probability that a borrower experiences an income shock large enough to miss a mortgage payment. Expected sign: positive.

House price index (Case-Shiller). Falling house prices push borrowers toward negative equity. A borrower who owes more than the property is worth has little ability to refinance out of financial distress and a weakened financial incentive to keep servicing the loan. Expected sign: negative.

Real GDP. GDP captures the broad macroeconomic environment that determines income growth, employment conditions, and consumer confidence simultaneously. Expected sign: negative.

10-year Treasury yield. Rising long term rates increase the cost of refinancing, which matters most for distressed borrowers trying to exit a loan they can no longer service. Higher rates reduce the cure rate for early stage delinquencies. Expected sign: positive.

High yield spread. Widening spreads signal broad financial market stress, tightening credit availability and raising the cost of any form of debt relief. Expected sign: positive.

For each series, four transforms are generated: quarter over quarter change (diff_1), year over year change (diff_4), quarter over quarter log growth (log_diff_1), and year over year log growth (log_diff_4). Each transform is then lagged at 0, 1, and 2 quarters, giving 12 candidate variables per series and a total universe of 60. Generating all transforms rather than pre-selecting means the data determines which frequency and horizon of a given series best predicts default, subject to the sign constraint from theory.

3.2 Stationarity Filter

We test all 60 candidates using the Augmented Dickey-Fuller test with AIC lag selection and drop any variable with p-value above 0.10 as a non-stationary regressor in a logistic model would produce spurious coefficient estimates. 33 of 60 variables pass this test. All 12 high yield spread variables return no ADF result due to insufficient data coverage in the macro panel and are excluded entirely.

3.3 Univariate Screen

We screen the 33 surviving variables for incremental predictive lift over a loan only baseline containing LTV, DTI, FICO, and loan age buckets. The screen runs on a 2 million

row random subsample rather than the full 50 million row panel, since fitting 33 separate models on the full dataset would be computationally prohibitive and 2 million rows produces stable AUC rankings. The baseline AUC on the subsample is 0.694. A variable passes if it adds AUC lift of at least 0.001 and its coefficient carries the correct sign for its canonical series, where the sign is set by economic theory at the series level and inherited by all transforms and lags. We argue that a variable that is predictive but carries the wrong sign is picking up a spurious correlation rather than the intended economic channel and is dropped regardless of lift. 16 variables pass this stage.

3.4 Correlation and Same Series Constraints

After screening for stationarity and sufficient AUC lift, we add two constraints to govern which three variable combinations are valid: (i) no two variables from the same canonical series can appear together, since including two transforms of unemployment, for example, adds collinearity without information from a distinct economic channel; (ii) all pairwise correlations within a combination must stay below 0.50. The correlation matrix across the 16 survivors shows several pairs above this threshold, most notably `diff` and `log_diff` transforms of the same series at matching lags reaching above 0.98, and unemployment and GDP variables at matching horizons reaching above 0.87. Starting from $C(16, 3) = 560$ combinations, the same series constraint reduces this to 50 and the correlation filter to 32 valid combinations.

3.5 Joint Evaluation and Final Selection

All 32 valid combinations are evaluated by fitting the full model on the 2 million row subsample and computing joint AUC. Any combination where a macro variable carries the wrong sign in the joint model is discarded even if AUC is high. A sign reversal in the joint model means the variable is behaving as a suppressor rather than an independent economic signal, which would make the model unreliable under stress scenarios where the macro paths are pushed far from historical norms. 22 of 32 combinations pass the joint sign check.

The selected combination produces the highest joint AUC lift among all sign-correct combinations:

unemployment_rate_log_diff_4_lag2. Year over year log change in the unemployment rate, lagged two quarters. The year over year horizon captures sustained labour market deterioration. The two quarter lag reflects the transmission delay between job loss and default: borrowers typically draw down savings and look for new work before missing a payment, so the default signal from a labour market shock takes time to materialise. Coefficient: +0.0125.

hpi_cs_log_diff_1_lag2. Quarter over quarter log change in the Case-Shiller index, lagged two quarters. The quarterly horizon captures recent price momentum rather than trend, which is what matters for equity erosion at the margin. The two quarter lag reflects that borrowers facing declining equity do not default immediately. They attempt refinancing, seek modifications, or draw on other resources first. Coefficient: -0.0183 .

real_gdp_log_diff_4. Year over year log change in real GDP, no lag. GDP enters without a lag because it reflects the conditions the borrower is already living through rather than a forward shock. A contracting economy reduces income growth, tightens labour markets, and weakens debt serviceability in ways that show up in default rates within the same quarter. Coefficient: -0.0306 .

The joint AUC lift of the selected combination over the loan only baseline is $+0.01750$, bringing combined AUC to 0.7115 on the screening subsample. Table 1 summarises the selection pipeline.

Table 1: Macro Variable Selection Pipeline

Stage	Variables remaining
Full universe	60
ADF pass ($p < 0.11$)	33
Univariate sign and lift screen	16
Valid three variable combinations	32
Joint sign correct	22
Selected	3

4. Model Specification and Estimation

4.1 Framework

The PD model is a discrete time logistic hazard model estimated by pooled logistic regression on the hazard panel constructed in Section 2. Each row asks the same conditional question: given that the loan is still performing at the end of quarter t , what is the probability it defaults in quarter $t + 1$. The panel’s discrete time structure with quarterly observations, time varying macro covariates, and an exit at first default means pooled logistic on this panel recovers the discrete time hazard likelihood directly without needing to specify a baseline hazard.

We choose logistic regression here for two reasons. First, it is more appropriate for a binary outcome than a linear probability model, which can produce predicted probabilities outside the unit interval and breaks down when quarterly conditional probabilities are chained into a lifetime projection. Second, it provides full interpretability: each coefficient carries a sign, a magnitude, and an economic rationale, and the effect of a macro shock

flows through directly in a way that can be traced and defended to a validator. More complex alternatives like gradient boosting would likely produce higher AUC but cannot offer that transparency in an IFRS 9 provisioning context.

4.2 Missing Value Treatment

Rows with missing values in any of LTV, DTI, FICO, or loan age are dropped before estimation. DTI has the largest missingness at 1,774,046 rows, followed by loan age at 1,220,963 and FICO at 333,050. The panel reduces from 54,199,536 to 50,933,331 rows. The default rate among dropped rows is 1.9987% against 2.7039% in the kept sample, meaning loans with missing characteristics tend to be lower risk on average and the drop is not missing at random with respect to the outcome.

4.3 Case Control Sampling and Intercept Correction

Even after quarterly aggregation the default rate in the estimation sample sits at 2.66% against a population rate of 0.27%. At that level of class imbalance the likelihood surface flattens and gradient based optimisers struggle to find meaningful separation. We address this through case control sampling at the loan level within each origination cohort. Every loan that ever defaults is retained in full with its complete quarterly history. From the never defaulting loans within each cohort we draw a random 5% sample, again retaining full histories. Sampling at the loan level rather than the row level preserves the longitudinal structure of each loan’s observation history, since row level sampling would produce a panel where a loan’s early quarters appear without its later ones and the time varying covariates lose their meaning.

Case control sampling keeps the slope coefficients consistent but shifts the intercept upward because the model sees a sample default rate far above the true population rate. We correct this by adding $\log(\tau_{eff})$ to the estimated intercept after fitting, where τ_{eff} is the effective non-event sampling fraction computed empirically as:

$$\tau_{eff} = \frac{\text{non-event rows in sample}}{\text{non-event rows in population}}$$

This is computed on the full panel before the train test split. In our case $\tau_{eff} = 0.098659$, giving a correction of $\log(0.098659) = -2.316088$, and the corrected intercept is:

$$\alpha^* = \hat{\alpha} + \log(\tau_{eff}) = -1.760730 + (-2.316088) = -4.076818$$

The alternative is the Weighted Exogenous Sample Maximum Likelihood estimator, which corrects all parameters simultaneously by weighting each observation by the inverse of its sampling probability. We use the intercept correction for simplicity since under case

control sampling the slope coefficients are consistent without weighting and only the intercept requires adjustment.

4.4 Model Specification

The log likelihood maximised over the training sample is the standard logistic:

$$\ell(\beta) = \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

where $p_i = \frac{1}{1+e^{-x_i^\top \beta}}$ and x_i contains a constant, LTV, DTI, FICO, the three selected macro variables, and five loan age dummy variables for buckets 6 to 12, 12 to 24, 24 to 60, 60 to 120, and 120 or more quarters, with the 0 to 6 quarter bucket as the reference category.

4.5 Clustered Standard Errors

The panel contains multiple observations per loan, so residuals are correlated within loan by construction. Without clustering, standard errors would be too narrow and z-statistics inflated. Standard errors are therefore clustered at the loan level. All 11 coefficients are significant at conventional levels after clustering.

5. Model Results

5.1 Discrimination

Table 2: Model Discrimination Metrics

Metric	Train	Test
AUC	0.7215	0.6546
Gini	0.4430	0.3092
KS	0.3301	0.2323
PR-AUC	0.0597	0.0440
Mean PD	0.2795%	0.2608%

The model achieves a train Gini of 0.44 and KS of 0.33, reasonable for a prime conforming portfolio where the borrower population is deliberately homogeneous. On the test sample Gini comes in at 0.31 and KS at 0.23. The test window includes the COVID period and a post-2022 rate environment the model was not trained on. We would not read the gap as model deterioration.

5.2 Calibration

Across most of the score distribution the model tracks realised default rates well. At the top end it underpredicts, with the highest risk loans defaulting at rates above what the

model assigns. For provisioning this means our ECL estimates on the riskiest loans are conservative. Test calibration is tighter than train. COVID compressed realised default rates in that window.

5.3 Coefficient Signs and Magnitudes

All 11 coefficients carry the expected sign and remain significant after loan level clustering. Among the loan characteristics FICO dominates on statistical precision with a z-statistic of -421 , reflecting how powerfully origination credit quality separates borrowers across 50 million rows. A 50 point move in FICO shifts log odds by 0.36. LTV and DTI come in nearly identical at 0.017, so a 10 point deterioration in either raises default odds by around 18%. On the macro side GDP carries the largest coefficient at -0.031 , followed by HPI at -0.018 and unemployment at 0.013. GDP picks up the broadest economic signal while HPI and unemployment operate through the equity and income channels more specifically.

The age dummy pattern is non-linear in a way the seasoning literature would predict but is worth walking through. The coefficient rises steeply from zero to 0.66 at 6 to 12 quarters and 1.15 at 12 to 24, where weaker borrowers start running into payment difficulties in the early years. It then flattens through 24 to 120 quarters, only 0.09 separating those two buckets as the pool works through its higher risk loans. At 120 or more quarters the coefficient picks back up to 1.72. Loans on the books for over a decade without defaulting have survived multiple stress cycles and what remains in that pool carries unobserved risk that origination characteristics alone do not capture.

6. Macro Scenario Model

6.1 Overview

The PD model takes three macro variables as inputs: unemployment YoY log growth lagged two quarters, HPI QoQ log growth lagged two quarters, and GDP YoY log growth. For IFRS 9 lifetime ECL we need quarterly paths for all three over the remaining life of each loan, under alternative economic conditions, so we estimate a Bayesian VAR on 152 quarters from 1987Q4 through 2025Q3 to generate these paths.

The BVAR is estimated on six variables rather than just the three PD model inputs. Adding fed funds rate QoQ change, 30 year mortgage rate QoQ change, and the 10y minus 2y Treasury spread gives the system enough structure to capture how monetary policy conditions propagate into labour markets and house prices over time, which a three variable system would miss. The three additional variables do not enter the PD model and are not exported; they serve only to condition the system dynamics.

All six variables enter as stationary transforms. A VAR on non-stationary series produces a companion matrix with eigenvalue above one and forecasts that diverge, so working in transforms is a requirement for stability. The transforms also happen to be exactly what the PD model needs as inputs, so the BVAR output feeds directly into the ECL projection with no further processing.

6.2 Estimation

We use the Minnesota prior with a Normal-Inverse-Wishart conjugate, which yields a closed form posterior and avoids MCMC. Each variable is assumed to follow a random walk by default, so own lag one coefficients are centred at one and cross lag coefficients at zero. Prior variance decays as $1/l^\alpha$ for lag l , with tightness $\lambda = 0.15$ and decay $\alpha = 2.0$, so more distant lags get shrunk harder toward zero. Lag order is selected by BIC, capped at $\min(6, T/n - 2)$ to prevent a singular residual covariance matrix, and BIC selects $p = 3$.

After fitting we check stability by verifying all eigenvalues of the companion matrix lie inside the unit circle, tightening λ if needed. At $\lambda = 0.15$ and $p = 3$ the maximum eigenvalue is 0.882. In sample R^2 runs at 0.86 for unemployment, 0.64 for GDP, and 0.50 for HPI, and against a random walk benchmark the BVAR wins on GDP across all forecast horizons and on HPI and the yield curve spread at short horizons.

As a sign check on the estimated dynamics, a positive unemployment shock produces a negative HPI growth response peaking at -0.09 percentage points, and a positive HPI shock produces a positive unemployment response peaking at $+0.74$, both in the expected direction.

6.3 Scenario Design

A plain BVAR mean reverts to its unconditional mean, which is a poor long run anchor for a 30 year projection since the long run value a variable settles at matters as much as the near term path for lifetime ECL. We therefore replace the unconditional mean with structural anchors derived from economic fundamentals. Unemployment anchors at zero YoY log growth, consistent with the economy operating at NAIRU. HPI anchors at 0.74 percentage points QoQ log growth, equivalent to roughly 3% annual appreciation given 2.1% inflation and around 1% real income growth. GDP anchors at 2.0% YoY, consistent with long run potential. Fed funds and mortgage rate changes anchor at zero and the yield curve spread at 1.5 percentage points.

Each scenario path runs 120 quarters and has three phases. In phase one the BVAR propagates shocks over 12 quarters using a triangular profile where the shock ramps up linearly to its peak, holds, then decays back to zero. In phase two a cosine bridge moves the path from the BVAR endpoint to the long run anchor over 16 to 24 quarters depending

on scenario severity, with zero slope at both endpoints so there is no kink in the transition. Phase three sits at the long run anchor for the remaining quarters.

In the stressed scenarios we also shift the long run anchors permanently after the shock. The mild scenario moves the HPI anchor down 0.05 percentage points QoQ and the severe scenario moves it down 0.12 percentage points and shifts the yield curve spread up 0.2 percentage points, on the basis that severe downturns tend to leave lasting damage to house prices and credit conditions.

Uncertainty bands come from 300 posterior predictive draws. The empirical 90% coverage over the calibration window came in at 45%, so we apply a scale factor of 2.0 to widen the bands to the target coverage.

6.4 Scenario Calibration

Three scenarios are generated. The baseline runs no shock and lets the BVAR evolve from current 2025Q3 conditions toward the long run anchors. The mild recession is calibrated to the 2001 episode: at peak, unemployment YoY log growth rises 1.5 percentage points, HPI QoQ growth falls 0.75 percentage points, and GDP YoY growth falls 1.0 percentage point. The severe stress is calibrated to the GFC, with unemployment rising 4.0 percentage points, HPI falling 2.0 percentage points QoQ, GDP falling 3.0 percentage points, and fed funds rising 1.0 percentage point.

One thing to note for the ECL results in Section 7 is that the baseline starts from 2025Q3 conditions where unemployment log growth is currently elevated, so even the no-shock scenario carries meaningful near term stress before the path mean reverts.

7. ECL Projection

This section applies the PD framework to a 2025Q3 new vintage portfolio. The primary output of this model is PD. The ECL calculation that follows is an illustrative exercise to examine what the PD estimates imply for provisioning under simple assumptions for LGD and EAD. A full ECL model would require dynamic LGD conditional on LTV and the HPI scenario path, and a more rigorous EAD treatment.

7.1 Portfolio

We assume a reporting date of 2025Q3 and a new origination vintage entering the portfolio at that point. The 2025Q3 Fannie Mae performance file serves as a proxy for this incoming vintage. Since we are interested in the credit risk of newly originated loans, we take only loans at age 3 months, corresponding to their first quarterly observation, and treat this as the portfolio snapshot at the reporting date. After dropping loans with missing

characteristics and filtering to LTV at or below 97% and DTI at or below 65% to stay within the conforming guidelines the model was trained on, we have 24,136 loans with mean FICO of 757, mean LTV of 75, and mean DTI of 38.

7.2 Projection Mechanics

For each loan and each scenario we chain the one quarter ahead conditional PD from the logit model forward over the remaining loan life, up to 120 quarters. At each step the model takes current loan characteristics and the macro values for that quarter from the scenario paths. Loan age advances by one quarter at each step and the macro inputs update from the scenario file. Loan characteristics other than age are held fixed at origination values, as discussed in Section 2.

The lifetime PD for loan i under scenario s is:

$$PD_{i,s}^{lifetime} = \sum_{t=1}^{T_i} h_{i,s,t} \cdot S_{i,s,t-1}$$

where $h_{i,s,t}$ is the conditional default probability at quarter t and $S_{i,s,t-1} = \prod_{j=1}^{t-1} (1 - h_{i,s,j})$ is the survival probability to the start of that quarter. The 12 month PD uses the same formula restricted to $t \leq 4$.

The scenario weighted PD is:

$$PD_i^{weighted} = \sum_s w_s \cdot PD_{i,s}^{lifetime}$$

with weights $w_{baseline} = 0.50$, $w_{mild} = 0.30$, $w_{severe} = 0.20$, and ECL as:

$$ECL = PD^{weighted} \times LGD \times EAD$$

We set LGD at 40% flat for several reasons. First, a proper mortgage LGD is a function of LTV at default and the HPI scenario path, since recovery depends on what the property is worth relative to the outstanding balance net of foreclosure costs. Computing this requires projecting LTV dynamically under each scenario, which is beyond the scope of this framework. Second, 40% is a conservative and commonly used stress testing assumption for US conforming mortgages that is defensible to a validator.

We set EAD to 1 because at age 3 months almost no principal has been paid down, so outstanding balance is essentially the origination amount. Normalising to 1 means ECL is expressed as a rate, which generalises across portfolio sizes and scales to any actual balance in production.

7.3 Results

Table 3: ECL Projection Results

Metric	Baseline	Mild	Severe	Weighted
Mean 12m PD	0.33%	0.36%	0.38%	0.35%
Mean Lifetime PD	26.30%	27.07%	28.00%	26.87%
Mean Lifetime ECL	—	—	—	10.75%

At the 12 month horizon the three scenarios produce similar results, weighted mean PD of 0.35% with only 5 basis points separating baseline from severe. The near term macro paths have not diverged much at that point, and a prime portfolio at age 3 months has had little time to season. At the lifetime horizon the scenario spread is more meaningful, with baseline at 26.3% and severe at 28.0% as the stressed macro paths and post-shock scarring push conditional default probabilities higher over 30 years. The weighted lifetime ECL of 10.75% is partly a function of the projection horizon and partly of where the macro environment sits in 2025Q3. As discussed in Section 6, the baseline itself carries near term stress before mean reverting, so a portfolio entering in a more benign environment would produce a lower lifetime ECL under the same framework.

8. Limitations and Future Work

The most fundamental limitation is that we trained the model on 2000 through 2014 originations, which does not include the period where mortgage rates moved from 3% to 7% in 18 months in 2022 to 2023, so the model has limited signal for how default behaves with key risk drivers during that environment. We could address this by simply retraining on post-2015 data.

We also do not model prepayment. We treat every loan as active for its full remaining contractual term, which overstates lifetime exposure in low rate environments where borrowers stay locked in and understates it in high rate ones where prepayment accelerates. We could extend the framework to a competing risks model that separates default and prepayment hazards.

As discussed in Section 2, we fix loan characteristics at origination to keep the projection tractable. We could update LTV quarterly using the amortisation schedule and the HPI scenario paths, which would improve PD prediction and also enable a dynamic LGD model. Related to this, we use only origination characteristics and macro conditions, with no behavioural signals observed during the loan life. Payment history is typically among the most important variables for 12 month PD and for Stage 2 and Stage 3 loans, and including behavioural variables would improve near term PD accuracy meaningfully.

On the macro side, the 50/30/20 scenario weights we use are judgment based rather than derived from a probability model. IFRS 9 requires probability weighted outcomes with documented rationale, and we could quantify ECL sensitivity to alternative weight assumptions as a next step. As discussed in Section 6, the BVAR baseline starts from elevated 2025Q3 macro conditions before mean reverting, so lifetime PD is higher than it would be at a neutral point in the cycle.

The ECL numbers in Section 7 are illustrative and sit on three simplifying assumptions. We set LGD at 40% flat, which does not capture how falling house prices under stress push LGD higher alongside PD, and we could extend this to a dynamic LTV-conditional LGD model. We normalise EAD to 1, reasonable for a new vintage at age 3 months, though a more complete treatment would account for balance rundown at later default dates. The scenario weights carry through from the PD projection into ECL and we have not quantified their sensitivity here.